

LAB 1

Introduction to Python Libraries

Python

- In the last few years there is an increasing community that creates **Data Mining tools in Python**
 - Python is overwhelmingly used today for data science tasks
 - It is also heavily used in industry
 - We will use Python for this class.
- There are tons of resources online for Python.
- For an introduction you can also look at the slides of the [Introduction to Programming](#) course by prof. N. Mamoulis
- I assume you have installed Python to your laptop by now and you have a good knowledge of programming in python.

Anaconda

- Installing libraries in Python can be complicated, so you should download the **Anaconda Scientific Python** distribution which will install most of the libraries that we will use.
 - Use Python 3.0
- Installing Anaconda installs a lot of libraries and also:
 - Anaconda Navigator
 - **Jupyter Notebook**: An interactive web-based interface for running python.
 - Anaconda Powershell: terminal for running commands

Jupyter Notebook

- Installing Anaconda will also install Jupyter Notebook.
- If you wish to install it in a different way, together with the relevant libraries you are free to do so.
- We will use Notebook for our examples and it is **required** for the assignments.
 - In almost all assignments you are required to submit a Notebook.

The Anaconda Navigator

Anaconda Navigator

File Help

 ANACONDA NAVIGATOR

Sign in to Anaconda Cloud

 Home

 Environments

 Learning

 Community

Documentation

Developer Blog



Applications on

base (root)

Channels

Refresh



JupyterLab

 1.0.2

An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.

Launch



Notebook

 6.0.0

Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.

Launch

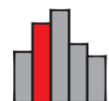


Spyder

3.3.6

Scientific PYTHON Development EnviRonment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features

Launch



Glueviz

0.15.2



Orange 3

3.23.1



RStudio

1.1.456

File

Paste

Clipboard

5

6

7

8

9

10

Slide 7 of 10

All Apps Documents Email Web More

Feedback

Best match

Anaconda Navigator (Anaconda3)
App

Apps

Anaconda Prompt (Anaconda3)

Jupyter Notebook (Anaconda3)

Anaconda Powershell Prompt (Anaconda3)

Search the web

ana - See web results

Folders (1+)

Jupyter Notebook (Anaconda3)
App

Open

Run as administrator

Open file location

Pin to Start

Pin to taskbar

Uninstall

Jupyter Notebook (Anaconda3)

Home Page - Select or c X

localhost:8889/tree

QuitLogout

FilesRunningClusters

Select items to perform actions on them.

0

NameLast ModifiedFile size

☐	3D Objects	21 days ago	
☐	Contacts	21 days ago	
☐	Documents	3 months ago	
☐	Downloads	4 hours ago	
☐	Dropbox (Personal)	8 days ago	
☐	Dropbox (Uol)	2 months ago	
☐	Favorites	10 days ago	
☐	Google Drive	3 days ago	
☐	Links	21 days ago	
☐	Music	21 days ago	
☐	OneDrive	2 days ago	
☐	Roaming	5 months ago	
☐	Saved Games	21 days ago	
☐	Searches	21 days ago	

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All Apps Documents Email Web More ▾ Feedback ...

Best match

 **Anaconda Powershell Prompt (Anaconda3)**
App

 **Anaconda Navigator (Anaconda3)** >
App

Apps

 **Anaconda Prompt (Anaconda3)** >

 **Jupyter Notebook (Anaconda3)** >

 **Spyder (Anaconda3)** >

 **Reset Spyder Settings (Anaconda3)** >

Search the web

 **anac** - See web results >



Anaconda Powershell Prompt (Anaconda3)
App

Open

Run as Administrator

Run ISE as Administrator

Windows PowerShell ISE

0 \$

Anaconda Powershell Prompt (Anaconda3)



```
Anaconda Powershell Prompt (Anaconda3)
(base) PS C:\Users\tsapa>
```

Installing Packages

- You can install packages from the Anaconda terminal using the command:
 - `conda install <name of package>`
- For example, [Seaborn](#) is a package for Statistical Data Visualization.
 - `conda install seaborn`
- [panda-datareader](#) is a package for loading online datasets.
 - `conda install pandas-datareader`

Notebooks

- Jupyter Notebook offers an interactive web-based interface for running code.
- The Notebook runs inside a browser.
- It allows you to interact with the code, running different parts of the code
- The results also appear in the browser, so you can have together the code and the results
- You can also add text, commenting on the results.
- We will now see some details on how to create notebooks

Changing the notebook default directory

- This used to be important before but now Jupyter Notebook takes you to the home directory
- From the Anaconda terminal type the command:
➤ `jupyter notebook --generate-config`
- This will generate `.jupyter/jupyter_notebook_config.py` file under your home directory.
- Find, un-comment and modify the line
`# c.NotebookApp.notebook_dir = ''` in the config file to point to the desired directory



Quit

Logout

Files

Running

Clusters

Nbextensions

Select items to perform actions on them.

Upload

New



0 /

Name

Notebook:

Python 3

Other:

Text File

Folder

Terminal

☐ 3D Objects☐ Contacts☐ Documents☐ Downloads☐ Dropbox (Personal)

10 months ago

☐ Dropbox (Uol)

2 months ago

☐ Favorites

8 months ago

☐ gensim-data

a year ago

☐ Google Drive

4 days ago

☐ IdeaProjects

18 days ago

☐ Links

8 months ago

☐ Music

8 months ago

☐ OneDrive

2 days ago

☐ OneDrive - ΠΑΝΕΠΙΣΤΗΜΙΟ ΙΩΑΝΝΙΝΩΝ

2 days ago

☐ Roaming

2 years ago

Home Page - Select or create a n x

Untitled1 - Jupyter Notebook x

localhost:8888/notebooks/Untitled1.ipynb?kernel_name=python3

Fair Information Ac... Data Mining

jupyter **Untitled1** Last Checkpoint: a few seconds ago (unsaved changes)

Logout

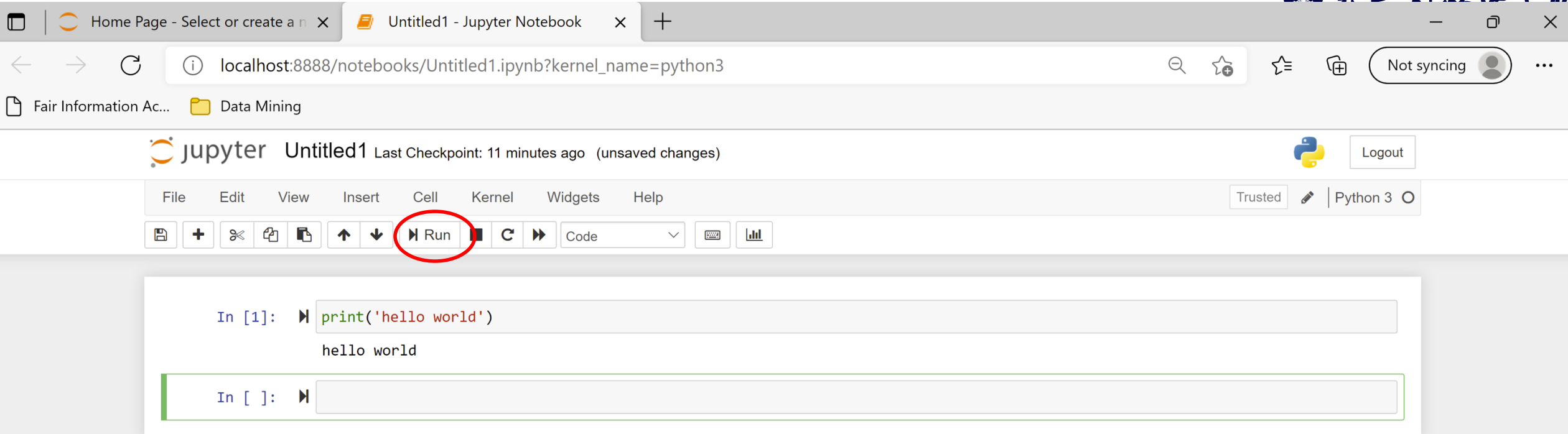
File Edit View Insert Cell Kernel Widgets Help

Trusted Python 3

Code

In []: |

- The notebook is organized in cells
- In each cell you can write either code or text
- The default behavior is code



Home Page - Select or create a n x

Untitled1 - Jupyter Notebook x

localhost:8888/notebooks/Untitled1.ipynb?kernel_name=python3

Fair Information Ac... Data Mining

jupyter Untitled1 Last Checkpoint: 11 minutes ago (unsaved changes)

Logout

File Edit View Insert Cell Kernel Widgets Help

Trusted Python 3

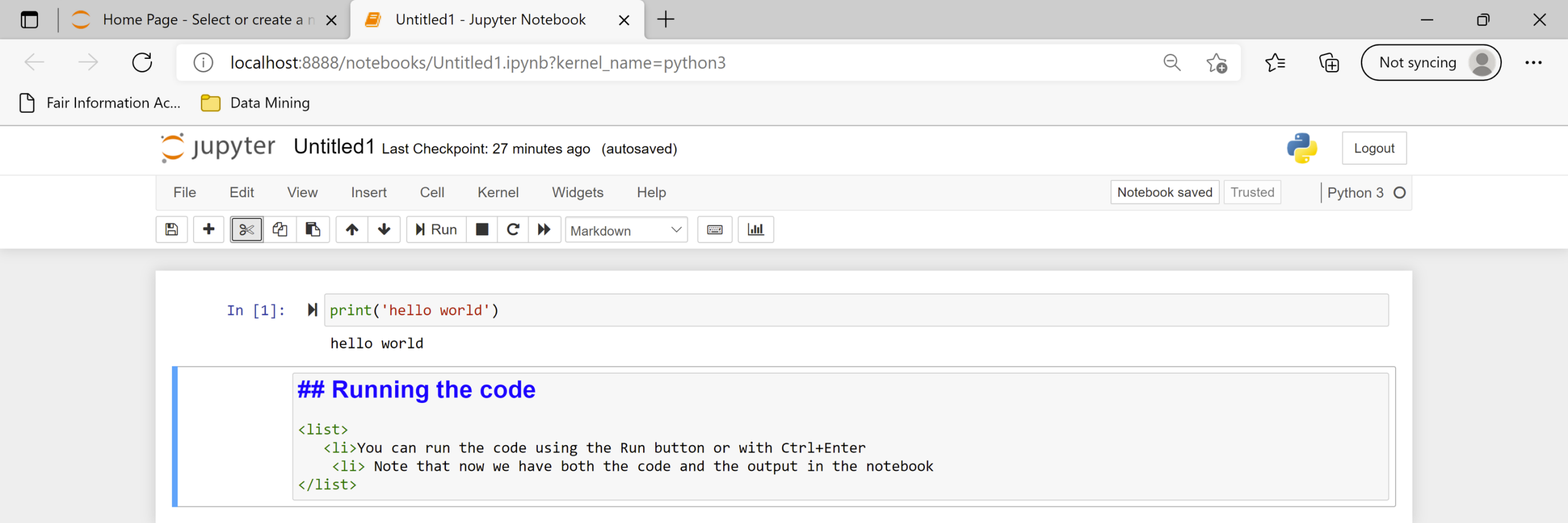
Run

```
In [1]: print('hello world')
```

hello world

```
In [ ]:
```

- You can run the code using the Run button or with Ctrl+Enter
- Note that now we have both the code and the output in the notebook



- You can also write text in Markdown language
 - You can combine HTML, and Latex, and there are some other commands
- You can learn more about Markdown by searching online, e.g.:
 - [Learn How to Write Markdown & LaTeX in The Jupyter Notebook | by Khelifi Ahmed Aziz | Towards Data Science](#)
- You need to Run the Markdown cell as well

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Fair Information Ac...Data Mining

jupyter

Untitled1

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Python

Logout

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Trusted

Python 3

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Code

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In [1]: ▶

```
print('hello world')
```

hello world

Running the code

- You can run the code using the Run button or with Ctrl+Enter
- Note that now we have both the code and the output in the notebook

In []: ▶

Home Page - Select or create a n x Untitled1 - Jupyter Notebook x +

localhost:8888/notebooks/Untitled1.ipynb?kernel_name=python3

Fair Information Ac... Data Mining

jupyter Untitled1 Last Checkpoint: 33 minutes ago (autosaved)

Logout

File Edit View Insert Cell Kernel Widgets Help

Trusted Python 3

New Notebook
Open...

Make a Copy...
Save as...
Rename...
Save and Checkpoint

Revert to Checkpoint

Print Preview

Download as

Trusted Notebook

Close and Halt

Run

Code

```
print('hello world')
```

hello world

Running the code

You can run the code using the Run button or with Ctrl+Enter

ate that now we have both the code and the output in the notebook

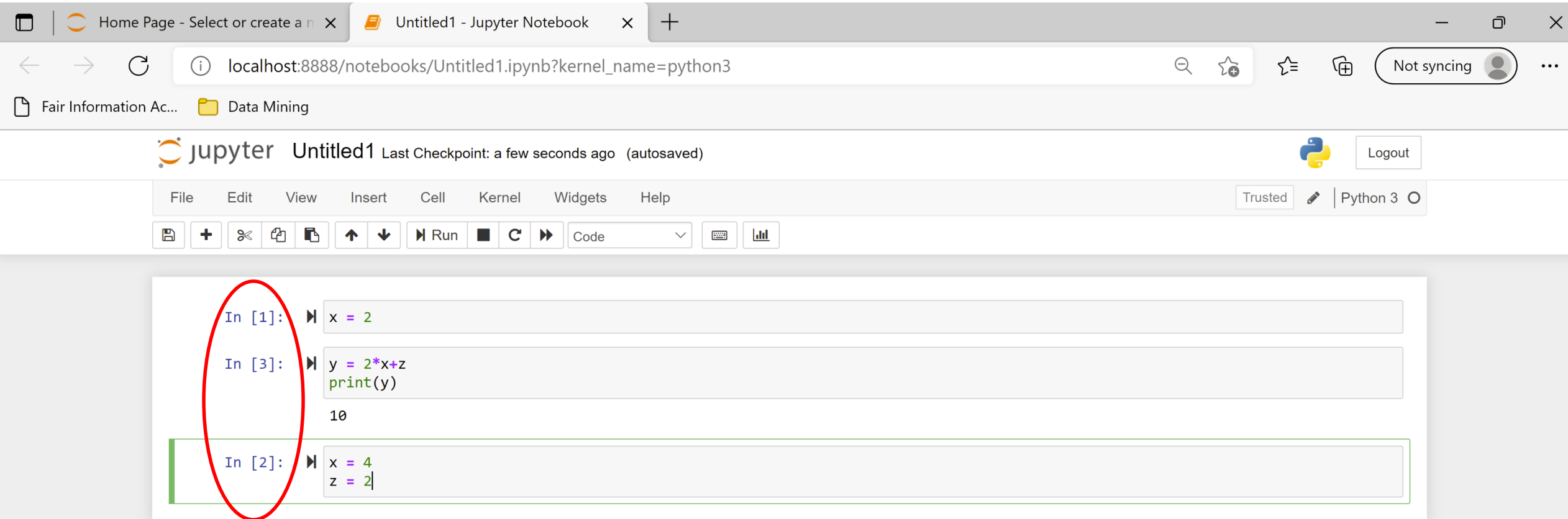
AsciiDoc (.asciidoc)
HTML (.html)
LaTeX (.tex)
Markdown (.md)
Notebook (.ipynb)
PDF via LaTeX (.pdf)
reST (.rst)
Python (.py)
Reveal.js slides (.slides.html)

- You can export the notebook into HTML or PDF

Attention!

- A notebook is run interactively, each time running a specific cell
- The state of the program remains in memory while the notebook is running
- Each cell has access to the current state of the memory
- You can jump between cells in a non-linear way
- You should be aware of the state of the memory of the notebook when you run a specific cell.

A simple example



The screenshot shows a Jupyter Notebook interface with three code cells. The first cell (In [1]:) contains the code `x = 2`. The second cell (In [3]:) contains the code `y = 2*x+z` and `print(y)`, with the output `10` displayed below it. The third cell (In [2]:) contains the code `x = 4` and `z = 2`. A red oval highlights the input prompts `In [1]:`, `In [3]:`, and `In [2]:` to show the execution order.

- The order in which the cells are executed is shown in the increasing numbers (not always useful)
- The second in order cell is executed third
 - So it has access to `z`, and uses the value 4 for `x`

Restarting

